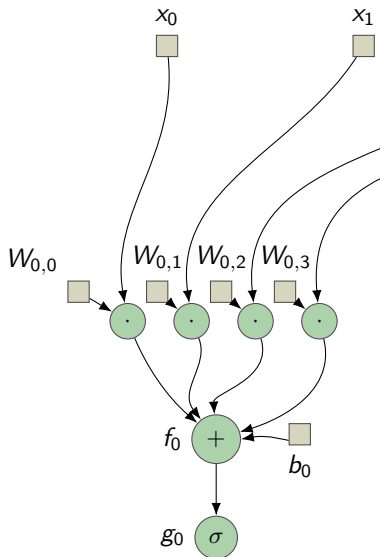


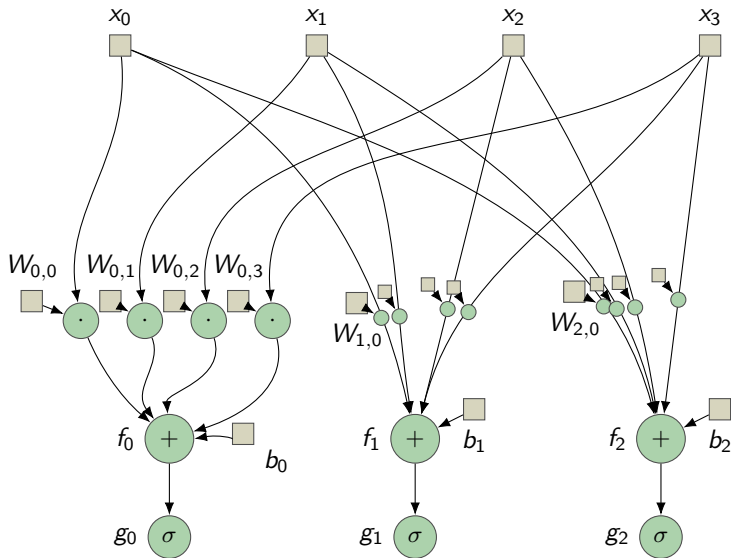
A neuron



An expression visualized as a tree

$$f_0 = W_{0,0} x_0 + \cdots + W_{0,N} x_N + b_0$$
$$g_0 = \sigma(f_0)$$

A layer of neurons



An expression with re-used values gives a DAG (directed acyclic graph).

A neural network

For computing gradients,
a neural network =
one big expression for the loss.

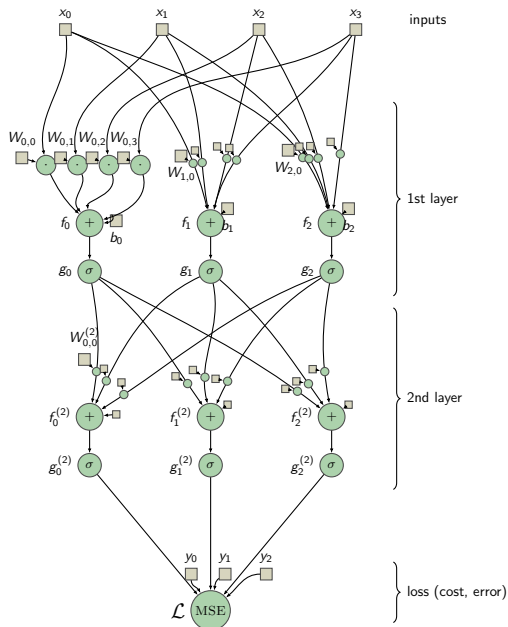
A function of:

inputs $\bar{x}$,

weights & biases

$W, \bar{b}, W^{(2)}, \bar{b}^{(2)}, \dots$,

and targets $\bar{y}$.



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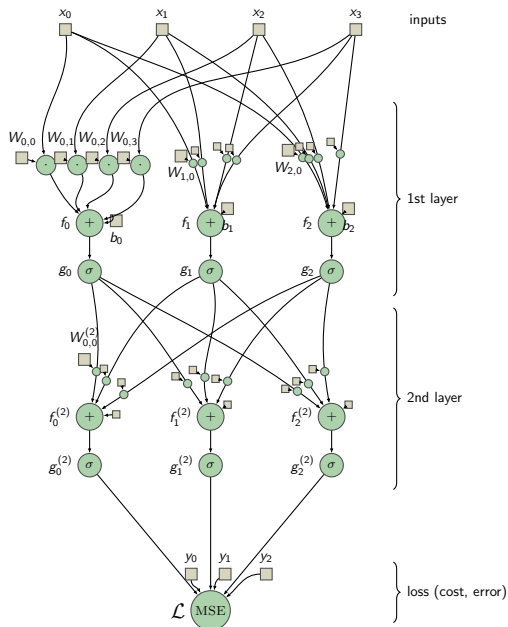
$W, \bar{b}, W^{(2)}, \bar{b}^{(2)}, \dots$,

and targets $\bar{y}$.

$$\mathcal{L} = \frac{1}{N^{(2)}} \sum_i (g_i^{(2)} - y_i)^2$$

where $g_i^{(2)} = \sigma(f_i^{(2)})$,

...



Gradients

Let $f(\bar{x})$ be a scalar function (like the loss).

The *gradient* at $\bar{x}$ is the vector pointing to where f most increases.

Computationally, it's just a vector of partial derivatives,

$$\nabla_{\bar{x}} f = \left(\frac{\partial f}{\partial x_0}, \dots, \frac{\partial f}{\partial x_{N-1}} \right)$$

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Computationally, it's just a vector of partial derivatives,

$$\nabla_{\bar{x}} f = \left(\frac{\partial f}{\partial x_0}, \dots, \frac{\partial f}{\partial x_{N-1}} \right)$$

If $\bar{x} \in \mathbb{R}^N$, then $\nabla_{\bar{x}} f \in \mathbb{R}^N$
(even if f has more inputs than just $\bar{x}$).

Derivates – chain rule

Let f be a function of x, y, z .

$$\frac{\partial f}{\partial x} = a(x, y, z) \quad \text{means} \quad \begin{array}{l} \text{increasing } x \text{ by } \varepsilon \\ \text{increases } f \text{ by } \sim a\varepsilon \quad (\text{for small } \varepsilon \in \mathbb{R}) \end{array}$$

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$$\text{Let } g = g(f(x, y, z)), \quad \frac{\partial f}{\partial x} = a_f, \quad \frac{\partial g}{\partial f} = a_g.$$

Increasing x by ε increases f by $a_f\varepsilon =: \varepsilon'$, which increases g by $a_g\varepsilon'$.

$$\text{So } \frac{\partial g}{\partial x} = a_g a_f.$$

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Note that $\frac{\partial g}{\partial f}$ depends on the value of f , which depends on x, y, z .

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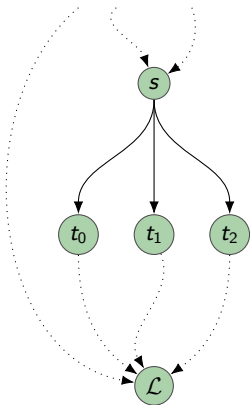
$$\text{So } \frac{\partial g}{\partial x} = a_g a_f. \quad \text{This proves } \frac{\partial g}{\partial x} = \frac{\partial g}{\partial f} \frac{\partial f}{\partial x}.$$

Note that $\frac{\partial g}{\partial f}$ depends on the value of f , which depends on x, y, z .

The expression for $\frac{\partial g}{\partial f}$ often involves subexpressions equal to g .

$$\text{Ex.: } g = \sigma(f) \implies \frac{\partial g}{\partial x} = \sigma(f)(1 - \sigma(f)) \frac{\partial f}{\partial x} = g(1 - g) \frac{\partial f}{\partial x}.$$

Derivates – in a graph

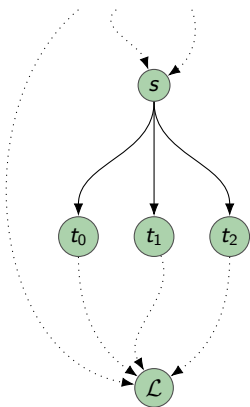


$$\frac{\partial \mathcal{L}}{\partial s} = \sum_{t \in \text{out}(s)} \frac{\partial \mathcal{L}}{\partial t} \cdot \frac{\partial t}{\partial s}$$

Because increasing s by ε
increases each t_i by $\frac{\partial t_i}{\partial s} \varepsilon$.

Each of these contribute to increasing $\mathcal{L}$
by $\frac{\partial \mathcal{L}}{\partial t_i} \left(\frac{\partial t_i}{\partial s} \varepsilon \right)$.

Derivates – in a graph



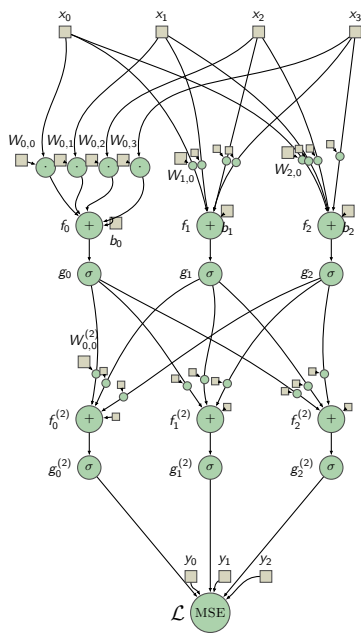
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Each of these contribute to increasing $\mathcal{L}$ by $\frac{\partial \mathcal{L}}{\partial t_i} \left(\frac{\partial t_i}{\partial s} \varepsilon \right)$.

So we can compute $\frac{\partial \mathcal{L}}{\partial s}$ for all nodes s , starting from $\frac{\partial \mathcal{L}}{\partial \mathcal{L}} = 1$ and going *back*, as long as we can compute how each out-neighbor t_i depends on s (that is, $\frac{\partial t_i}{\partial s}$).

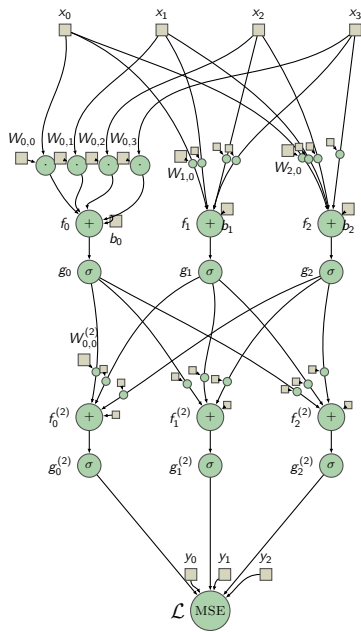
Forward pass



Input $\bar{x} \in \mathbb{R}^{N^{(0)}}$

$x_0, \dots, x_{N^{(0)}-1}$

Forward pass



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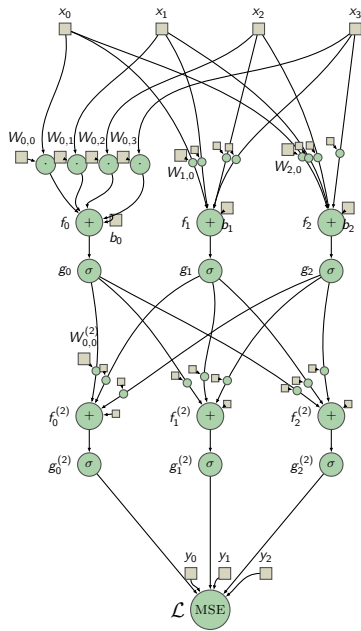
Weights $W^{(1)} \in \mathbb{R}^{N^{(1)} \times N^{(0)}}$ and biases $b^{(1)} \in \mathbb{R}^{N^{(1)}}$

Pre-activations

$$\bar{f}^{(1)} = W^{(1)}\bar{x} + \bar{b}^{(1)} \in \mathbb{R}^{N^{(1)}}$$

$$f_i^{(1)} = \sum_j W_{i,j}^{(1)} x_j + b_i^{(1)}$$

Forward pass



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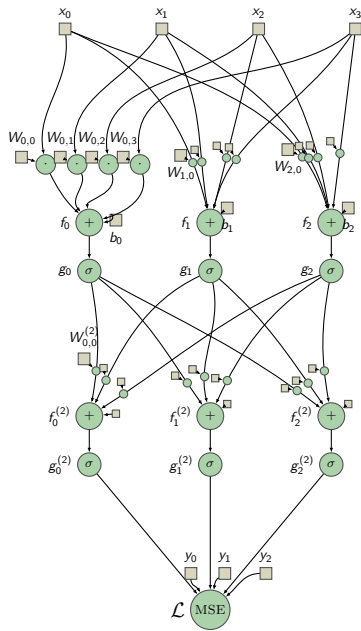
$$f_i^{(1)} = \sum_j W_{i,j}^{(1)} x_j + b_i^{(1)}$$

Activations

$$\bar{g}^{(1)} = \sigma(\bar{f}^{(1)})$$

$$g_i^{(1)} = \sigma(f_i^{(1)})$$

Forward pass



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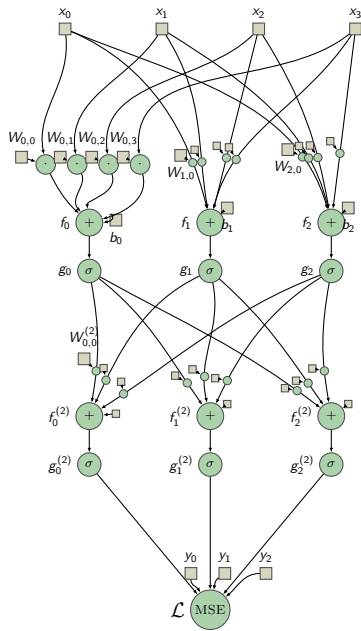
$$\bar{g}^{(1)} = \sigma(\bar{f}^{(1)})$$

$$g_i^{(1)} = \sigma(f_i^{(1)})$$

$$\bar{f}^{(2)} = W^{(2)}\bar{g}^{(1)} + \bar{b}^{(2)} \in \mathbb{R}^{N^{(2)}}$$

$$\bar{g}^{(2)} = \sigma(\bar{f}^{(2)})$$

Forward pass



Input $\bar{x} \in \mathbb{R}^{N^{(0)}}$

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Activations

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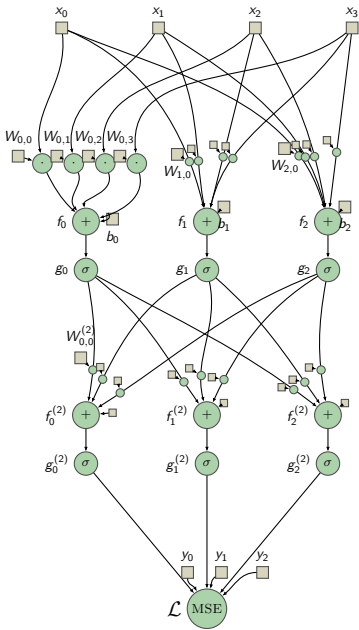
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$$\bar{g}^{(2)} = \sigma(\bar{f}^{(2)})$$

$$\mathcal{L} = \text{mean} \left[(\bar{g}^{(2)} - \bar{y})^2 \right] = \frac{1}{N^{(2)}} \sum_i (g_i^{(2)} - y_i)^2$$

Forward pass



Input $\bar{x} \in \mathbb{R}^{N^{(0)}}$

Weights $W^{(1)} \in \mathbb{R}^{N^{(1)} \times N^{(0)}}$ and biases $b^{(1)} \in \mathbb{R}^{N^{(1)}}$

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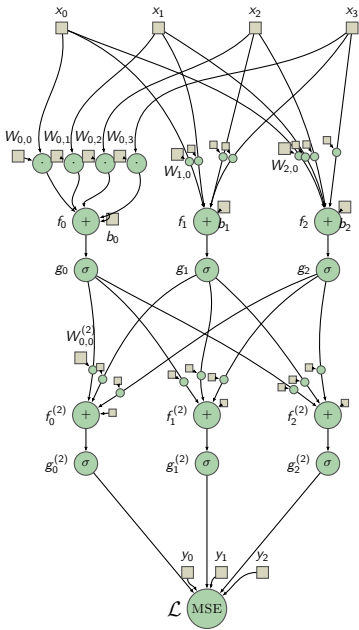
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$$\mathcal{L} = \text{mean} \left[(\bar{g}^{(2)} - \bar{y})^2 \right]$$

Forward pass – code



```
import numpy as np
```

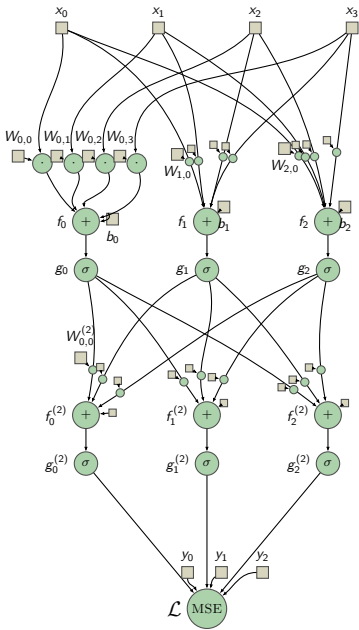
```
g = x
```

```
for b, w in zip(biases, weights):
```

```
    f = w @ g + b
```

```
    g = sigmoid(f)
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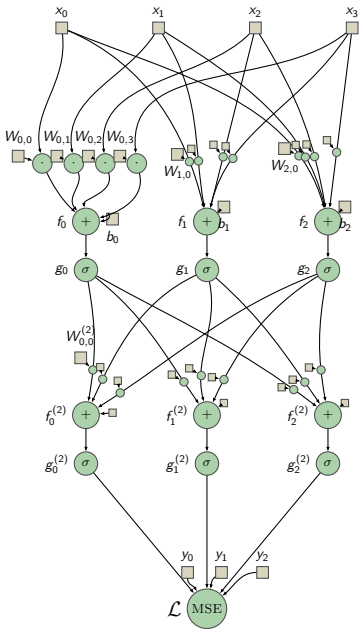
When batched:

x has shape $(B, N^{(0)})$

g has shape

before iteration i

Forward pass – code



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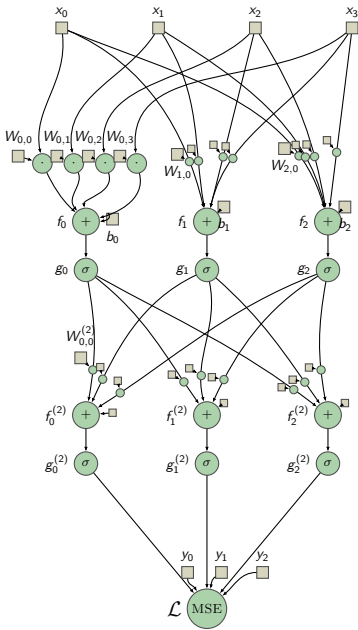
When batched:

x has shape $(B, N^{(0)})$

g has shape $(B, N^{(i)})$ before iteration i

w has shape

Forward pass – code



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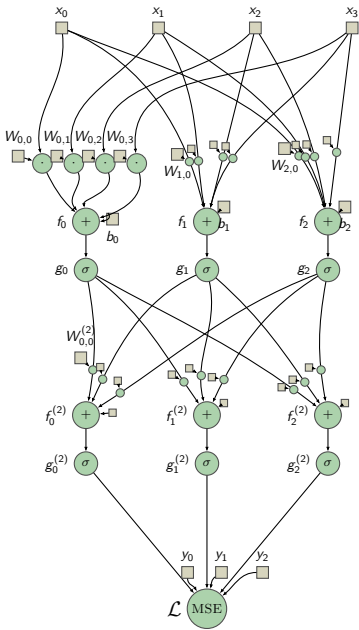
x has shape $(B, N^{(0)})$

g has shape $(B, N^{(i)})$ before iteration i

w has shape $(N^{(i+1)}, N^{(i)})$

$w @ g.T$ has shape

Forward pass – code



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for b, w in zip(biases, weights):
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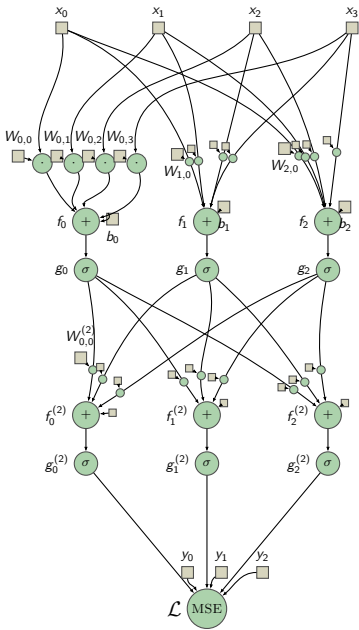
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$w @ g.T$ has shape $(N^{(i+1)}, B)$

Forward pass – code



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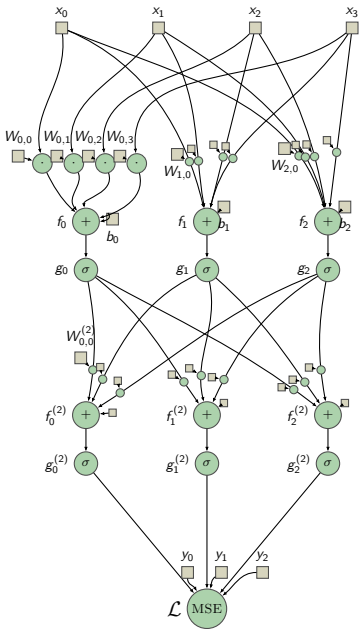
g has shape $(B, N^{(i)})$ before iteration i

w has shape $(N^{(i+1)}, N^{(i)})$

$w @ g.T$ has shape $(N^{(i+1)}, B)$

$(w @ g.T).T$ has shape $(B, N^{(i+1)})$

Forward pass – code



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When batched:

x has shape $(B, N^{(0)})$

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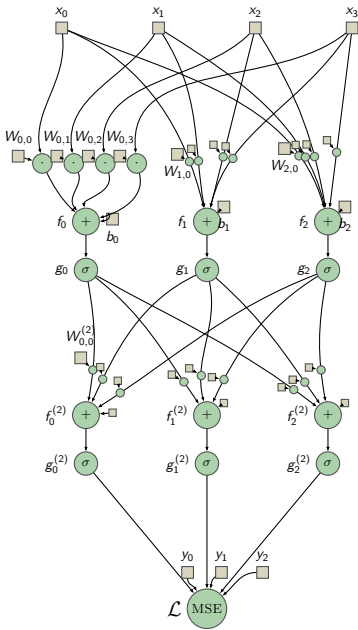
w has shape $(N^{(i+1)}, N^{(i)})$

$w @ g.T$ has shape $(N^{(i+1)}, B)$

$(w @ g.T).T$ has shape $(B, N^{(i+1)})$

b has shape

Forward pass – code



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When batched:

x has shape $(B, N^{(0)})$

g has shape $(B, N^{(i)})$ before iteration i

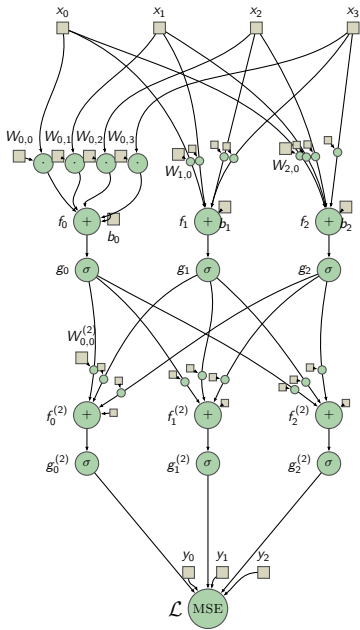
w has shape $(N^{(i+1)}, N^{(i)})$

$w @ g.T$ has shape $(N^{(i+1)}, B)$

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b has shape $(N^{(i+1)})$

Forward pass – code



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import numpy as np
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$$g = x$$

```
for b, w in zip(biases, weights):
```

$$\mathbf{f} = \mathbf{W} @ \mathbf{g} + \mathbf{b}$$
$$g = \text{sigmoid}(f)$$

When batched:

$\mathbf{x}$ has shape $(B, N^{(0)})$

$\mathbf{g}$ has shape $(B, N^{(i)})$ before iteration i

w has shape $(N^{(i+1)}, N^{(i)})$

$w @ g.T$ has shape $(N^{(i+1)}, B)$

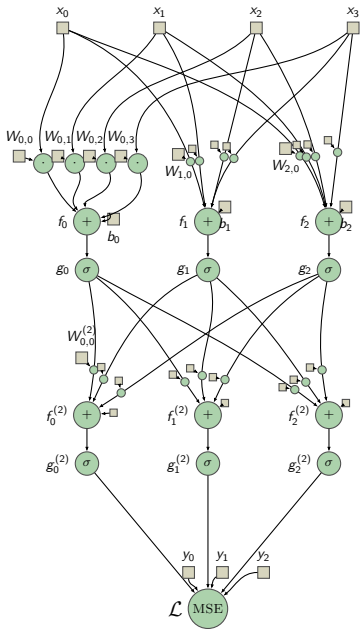
$(w @ g.T).T$ has shape $(B, N^{(i+1)})$

b has shape $(N^{(i+1)})$

$$\mathbf{f} = (\mathbf{w} @ \mathbf{g.T}).T + \mathbf{b}$$

(with `b` broadcasted along the batch)

Forward pass – code



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When batched:

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w has shape $(N^{(i+1)}, N^{(i)})$

$w @ g.T$ has shape $(N^{(i+1)}, B)$

$(w @ g.T).T$ has shape $(B, N^{(i+1)})$

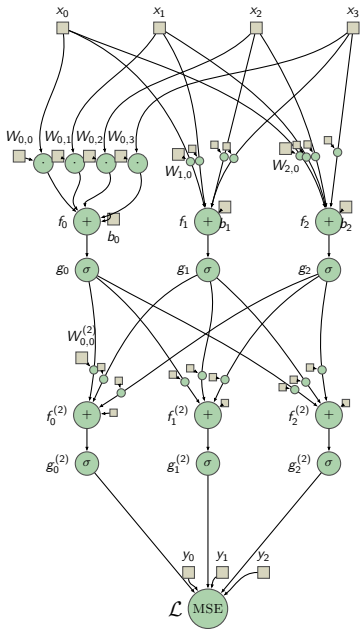
b has shape $(N^{(i+1)})$

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f = (w @ g.T).T + b
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(with b broadcasted along the batch)

@ is the same as `np.matmul`

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b has shape $(N^{(i+1)})$

```
f = (w @ g.T).T + b
```

(with b broadcasted along the batch)

@ is the same as `np.matmul`

```
loss = ((g - y) ** 2).mean()
```

over outputs and batch, $\frac{1}{B \cdot N^{(2)}}$

Backward pass

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} =$$

Backward pass

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right) \frac{\partial (g_i^{(L)} - y_i)}{\partial g_i^{(L)}}$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} =$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

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$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial \mathbf{g}_j^{(\ell-1)}} =$$

$$\mathbf{f}_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} \mathbf{g}_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{f}_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{g}_i^{(\ell)}} \sigma'(\mathbf{f}_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial \mathbf{g}_i^{(\ell)}} \mathbf{g}_i^{(\ell)} (1 - \mathbf{g}_i^{(\ell)})$$

$$\mathbf{g}_i^{(\ell)} = \sigma(\mathbf{f}_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{g}_i^{(L)}} = \frac{2}{N^{(L)}} \left(\mathbf{g}_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (\mathbf{g}_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial \mathbf{g}_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} \frac{\partial f_i^{(\ell)}}{\partial \mathbf{g}_j^{(\ell-1)}}$$

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

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$$\frac{\partial \mathcal{L}}{\partial \mathbf{g}_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial \mathbf{g}_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$f_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

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$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$f_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} \frac{\partial f_i^{(\ell)}}{\partial b_i^{(\ell)}}$$

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$f_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

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$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

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Backward pass

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$f_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} \frac{\partial f_i^{(\ell)}}{\partial W_{i,j}^{(\ell)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

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$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

$$f_i^{(\ell)} = \sum_j W_{i,j}^{(\ell)} g_j^{(\ell-1)} + b_i^{(\ell)}$$

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} \sigma'(f_i^{(\ell)}) = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$g_i^{(\ell)} = \sigma(f_i^{(\ell)})$$

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\mathcal{L} = \frac{1}{N^{(L)}} \sum_i (g_i^{(L)} - y_i)^2$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\vec{b}^{(\ell)}} \mathcal{L} = \nabla_{\vec{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{N^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} =$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\vec{b}^{(\ell)}} \mathcal{L} = \nabla_{\vec{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{N^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right) \frac{\partial (g_i^{(L)} - y_i)}{\partial g_i^{(L)}}$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = \quad \in \mathbb{R}^{M^{(\ell)} \times M^{(\ell-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\vec{b}^{(\ell)}} \mathcal{L} = \nabla_{\vec{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{M^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = (\nabla_{\tilde{f}^{(\ell)}} \mathcal{L})^\top \tilde{g}^{(\ell-1)} \in \mathbb{R}^{M^{(\ell)} \times M^{(\ell-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\tilde{b}^{(\ell)}} \mathcal{L} = \nabla_{\tilde{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{M^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial W_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = (\nabla_{\tilde{f}^{(\ell)}} \mathcal{L})^\top \tilde{g}^{(\ell-1)} \in \mathbb{R}^{M^{(\ell)} \times M^{(\ell-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\tilde{b}^{(\ell)}} \mathcal{L} = \nabla_{\tilde{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{M^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} W_{i,j}^{(\ell)}$$

$$\nabla_{\tilde{g}^{(\ell-1)}} \mathcal{L} = W^{(\ell)\top} (\nabla_{\tilde{f}^{(\ell)}} \mathcal{L})$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial w_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = (\nabla_{\tilde{f}^{(\ell)}} \mathcal{L})^\top \tilde{g}^{(\ell-1)} \in \mathbb{R}^{N^{(\ell)} \times N^{(\ell-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\tilde{b}^{(\ell)}} \mathcal{L} = \nabla_{\tilde{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{N^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$\nabla_{\tilde{g}^{(\ell-1)}} \mathcal{L} = W^{(\ell)\top} (\nabla_{\tilde{f}^{(\ell)}} \mathcal{L})$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$\nabla_{\tilde{f}^{(\ell)}} \mathcal{L} =$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} \left(g_i^{(L)} - y_i \right)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial w_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = (\nabla_{\vec{f}^{(\ell)}} \mathcal{L})^\top \vec{g}^{(\ell-1)} \in \mathbb{R}^{N^{(\ell)} \times N^{(\ell-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\vec{b}^{(\ell)}} \mathcal{L} = \nabla_{\vec{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{N^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$\nabla_{\vec{g}^{(\ell-1)}} \mathcal{L} = W^{(\ell)\top} (\nabla_{\vec{f}^{(\ell)}} \mathcal{L})$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$\nabla_{\vec{f}^{(\ell)}} \mathcal{L} = (\nabla_{\vec{g}^{(\ell)}} \mathcal{L}) \odot \vec{g}^{(\ell)} \odot (1 - \vec{g}^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{N^{(L)}} (g_i^{(L)} - y_i)$$

Backward pass

Weight and biases

$$\frac{\partial \mathcal{L}}{\partial w_{i,j}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} g_j^{(\ell-1)}$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = (\nabla_{\vec{f}^{(\ell)}} \mathcal{L})^\top \vec{g}^{(\ell-1)} \in \mathbb{R}^{M^{(\ell)} \times M^{(\ell-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}}$$

$$\nabla_{\vec{b}^{(\ell)}} \mathcal{L} = \nabla_{\vec{f}^{(\ell)}} \mathcal{L} \in \mathbb{R}^{M^{(\ell)}}$$

Activations

$$\frac{\partial \mathcal{L}}{\partial g_j^{(\ell-1)}} = \sum_i \frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} w_{i,j}^{(\ell)}$$

$$\nabla_{\vec{g}^{(\ell-1)}} \mathcal{L} = W^{(\ell)\top} (\nabla_{\vec{f}^{(\ell)}} \mathcal{L})$$

Pre-activations

$$\frac{\partial \mathcal{L}}{\partial f_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial g_i^{(\ell)}} g_i^{(\ell)} (1 - g_i^{(\ell)})$$

$$\nabla_{\vec{f}^{(\ell)}} \mathcal{L} = (\nabla_{\vec{g}^{(\ell)}} \mathcal{L}) \odot \vec{g}^{(\ell)} \odot (1 - \vec{g}^{(\ell)})$$

Output activations

$$\frac{\partial \mathcal{L}}{\partial g_i^{(L)}} = \frac{2}{M^{(L)}} \left(g_i^{(L)} - y_i \right)$$

$$\nabla_{\vec{g}^{(L)}} \mathcal{L} = \frac{2}{M^{(L)}} \left(\vec{g}^{(L)} - \vec{y} \right)$$